

Diego A. Lopez G.

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Education

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| PhD | University of Kentucky , Electrical Engineering <ul style="list-style-type: none"> • Advisor: Professor Dan M. Ionel • GPA: 4.0/4.0 | Lexington, KY
2029 |
| BS | Universidad de La Sabana , Mechanical Engineering <ul style="list-style-type: none"> • Excellence Scholarship recipient • GPA: 3.5/4.0 (equiv. excellent) | Chia, Colombia
Mar 2024 |

Experience

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| SPARK Lab, University of Kentucky , Research Assistant <ul style="list-style-type: none"> • Contributions to projects sponsored directly from NASA and industry • Employed Autodesk Inventor and Fusion 360 for 3D CAD modeling, manufacturing drawings, and assembly videos • Performed parametric electromagnetic modeling, analysis, and optimization of electric machines with Ansys Maxwell finite element analysis (FEA) • Operated lab equipment, including power sources, oscilloscopes and power analyzers • Designed mechanical fixtures for motor prototypes, combining off-the-shelf parts, in-house manufacturing, and external collaborators • Scripted and processed data using MATLAB and Python | Lexington, KY
Aug 2024 – present
2 years 3 months |
| Stanley and Karen Pigman College of Eng., University of Kentucky , Teaching Assistant <ul style="list-style-type: none"> • Guided engineering students from diverse backgrounds in understanding and navigating complex engineering concepts • <i>Electromechanics and Electromagnetics</i>: Strong theoretical foundations • <i>Electronic Packaging</i>: PCB design and manufacturing using KiCad • <i>First-Year Engineering</i>: Microcontroller-based projects, guiding students on how to code, design, and build using MATLAB, CAD software, and 3D printing | Lexington, KY
Aug 2024 – present
2 years 3 months |

Projects

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| Project Polaris <ul style="list-style-type: none"> • Designed and built a buoyant rover prototype for Titan exploration within an international student space program • Led an international, multidisciplinary team of 50+ members • Developed the instrumentation system for scientific payloads and for guidance and navigation, using Raspberry Pi and Arduino peripherals • Coauthored research papers and was selected as a speaker at the International Astronautical Congress (IAC) 2022 and AIAA SciTech 2024 | 2022 – 2023 |
| Ultra-Efficient EV <ul style="list-style-type: none"> • Designed and built an ultra-efficient battery-electric vehicle for Shell Eco-marathon competition • Developed the electronic speed controller of a BLDC motor, including schematic reading, PCB design in KiCad, and manual assembly • Received honorable mention for the innovative technical design of the vehicle at Shell Eco-Marathon Brazil 2022 • Performed 3D modeling in Autodesk Inventor • Manufactured carbon fiber, 3D-printed, and CNC-machined parts • Collaborated as a team player to solve engineering problems in a fast-paced environment | 2019 – 2024 |

Skills and Relevant Coursework

Soft skills: Excellent written and verbal communication, strong interpersonal skills with demonstrated effectiveness in a fast-paced team environment, can-do attitude, and results orientation

Software: Ansys Maxwell, MATLAB, Python, Autodesk CAD, KiCad, Excel, LaTeX, and Microsoft Office

Prototyping: 3D CAD modeling and manufacturing drawings, mechanical fixture design for motor prototypes, 3D printing, CNC machining, carbon fiber fabrication, PCB design and assembly, microcontroller-based systems

Ph.D. coursework: Electric Machines Computer Aided Eng. (A), Machine Learning for Motor Design and Analysis (A), Electromagnetic Fields (A), Electric Vehicles and Aircraft (A), Mechanical Design with Finite Element Methods (A)

Research Areas: Electric machine design and optimization, axial-flux motors, PCB windings, electric aircraft

Publications

PM-Free Axial-Flux AC Motor with Reluctance Rotor and Ferromagnetic-Core Stator with PCB DC Excitation for Renewable Energy Systems

Lopez-Guerrero, D. A., Vatani, M., Ionel, D. M.
(ICRERA 2026)

Halbach-Mallinson Array Rotor with Variable-Length PMs for Axial-Flux Machines

Lopez-Guerrero, D. A., Vatani, M., Goode-Kulchar, S., Ionel, D. M.
(ECCE 2026)

Fault Detection and Mitigation in a Multi-Stage Coreless Axial-Flux PM Motor System

Deponti, M., Vatani, M., **Lopez-Guerrero, D. A.**, Perini, R., Ionel, D. M.
(ECCE 2026)

Electrical Fault Tolerance of a Multi-Stage Axial-Flux PM Drivetrain for Electrified Propulsion

Deponti, M., Lewis, D. D., **Lopez-Guerrero, D. A.**, Vatani, M., Perini, R., Ionel, D. M.
[10.1109/ITECEATS66641.2026.11593010](#) (ITEC 2026)

Multi Electric Machines with Series and Parallel Electromechanical Combinations for Aircraft

Stewart, D. R., Lewis, D. D., Vatani, M., **Lopez-Guerrero, D. A.**, Ionel, D. M.
[10.1109/ECCE58356.2025.11260309](#) (ECCE 2025)

Mitigation of PM Eddy Current Losses Using Offset Windings in Coreless AFPM Machines

Vatani, M., **Lopez-Guerrero, D. A.**, Eastham, J. F., Ionel, D. M.
[10.1109/ITEC63604.2025.11098002](#) (ITEC 2025)

Selected Honors

- Best Student Demo Award, IEEE International Electric Machines & Drives Conference 2025
- Winner of the 2021 Sabanahack hackathon, awarded by Prodigious and Universidad de La Sabana
- Winner of the Excellence Scholarship awarded by the Universidad de La Sabana College of Engineering (2019–2023)
- Scored in the top 1% nationally on standardized exams for high school (2018) and university graduates (2023)
- Certified C1 English level (IELTS 7.5; Duolingo English Test 145)

References

Professor Dan M Ionel, PhD, MEng, IEEE Fellow

L. Stanley Pigman Chair in Power, University of Kentucky

Honorary Professor, University College London (UCL)

Director, [SPARK Lab](#) and [PEIK Institute](#)

University of Kentucky, dan.ionel@uky.edu